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PAPER_933: Extended AGN Three-Point Jet Power Curves

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 212 **Source:** agn_{jet_power_curves_extended}.py (ThreePointJetPowerCurve) **Calculator:** ExtendedAGNThreePointJetPowerCalc (CP4 #517) **CVW:** v2.0.0 compliant

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Abstract

We present explicit three-point jet power enhancement curves for 3C273 and TON618, computed at phonon linewidths Gamma = 0.05, 0.10, and 0.30 THz. Jet modulation via M_jet(Gamma) couples to the Blandford-Znajek base power P_BZ through the relation P_jet = P_BZ (1 + M_jet). For 3C273 (A_jet ~ 1.05), enhancements are 3.1/2.4/1.5x at the three Gamma points. For TON618 (A_jet ~ 1.40), enhancements reach 3.8/2.9/1.7x, reflecting the stronger phonon-jet coupling in ultra-massive BH systems.

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1. Core Equations

Blandford-Znajek power:

$$P_{\mathrm{BZ}} = \frac{B^2}{8\pi}\left(\frac{r_H}{c}\right)^2 a^2 c$$

Jet modulation factor:

$$M_{\mathrm{jet}}(\Gamma) = 1 + A_{\mathrm{jet}} \exp\left(-\frac{(\Gamma-\Gamma_0)^2}{2\sigma_G^2}\right)$$

where $\Gamma_0 = 2\pi \times 0.1 \times 10^{12}$ rad/s and $\sigma_G = 0.08 \times 2\pi \times 10^{12}$ rad/s.

Total jet power:

$$P_{\mathrm{jet}} = P_{\mathrm{BZ}}(1 + M_{\mathrm{jet}})$$

Enhancement ratio:

$$\mathrm{Enhancement} = \frac{P_{\mathrm{jet}}(\Gamma)}{P_{\mathrm{BZ}}}$$

Three-Point Results

System	Gamma=0.05	Gamma=0.10	Gamma=0.30
3C273 (A=1.05)	3.1x	2.4x	1.5x
TON618 (A=1.40)	3.8x	2.9x	1.7x

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2. UQFF Integration

The `ExtendedAGNThreePointJetPowerCalc` (CP4 #517) accepts M_Msun, a_spin, B_T, and A_jet as parameters. The simulate() method sweeps A_jet through [0.8, 1.05, 1.40, 2.0] to map how coupling strength affects the three-point enhancement profile.

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3. Source Data

- **File:** agn_{jet_power_curves_extended}.py
- **Session:** 212
- **CP4 Class:** ExtendedAGNThreePointJetPowerCalc (#517)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. Blandford, R.D. & Znajek, R.L. — Electromagnetic extraction of energy from Kerr black holes

(1977)

3. Ghisellini, G. et al. — General physical properties of bright Fermi blazars (2010)

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_pc/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ kg/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\rm BH} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\rm SCm}/\omega_{\rm bulge})$.

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4}\,\mathrm{day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	$\omega_{\rm SCm}$	$2\pi \times 1.25$ THz	Phonon resonance
SCm vacuum density	$\rho_{\rm SCm}$	$7.09 \times 10^{-37}\,\mathrm{kg/m^3}$	Fundamental

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency $\omega_{\rm SCm}$	$2\pi \times 1.25$ THz (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
Vacuum energy $\rho_{\rm vac}$	7.09×10^{-37} kg/m^3	$\rho_{\rm vac} \sim 10^{-29}$ g/cm^3	Planck 2018	Novel SCm scale
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator).

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: SCm-phonon (lattice resonance)

§A.2 Lagrangian Density

$$\mathcal{L}_{SCm\_phonon} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\mathrm{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial(\partial_m u \phi)} = 0 \implies F_{U,Bi\_i} = -\nabla U_{\rm eff} + \Phi \cdot S_{26} \cdot E_{\rm net}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ lattice resonance $\rightarrow$ F_{U,Bi_i} unified force $\rightarrow$ observational prediction

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§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$\text{VDS} = \rho_{\text{SCm}} \cdot S_{26} \cdot \Phi_{1.25\text{THz}}/\Phi_0$$

VDS sub-ratio: 0.1

§B.2 Dipole Vortex Primes (DVP)

DVP prime: 2 (sub-threshold)

§B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: 10^4 yr

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy

14 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
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